GenAI for Engineers Lesson Plan

IES | v3.1 | LP-aligned one-day delivery

Follow the supplied Lesson Plan with Hands-On Labs (v2 - PUB Feedback) module order and clock times. Retain its six modules; consolidate learner work into only 65 minutes of guided Excel exercises within Module 3. Other practical examples are trainer-led demonstrations or discussion. The original agent-build outcomes are demonstrated, not assessed as independent learner deployment.

09:00–17:00 = 480 minutes: 390 teaching minutes + 60 lunch + 30 tea. The source LP calls this seven training hours excluding lunch; that includes tea. This delivery plan uses 6.5 net teaching hours. Guided Excel time is 25 + 20 + 20 = 65 minutes, included within Module 3, never added on top.

Time	LP module	Delivery allocation
09:00–10:00	1 · Copilot ecosystem	15 overview; 10 access check; 20 Excel demonstration; 15 discussion
10:00–11:15	2 · Prompting for engineering documents	20 RISE/E-MAP; 20 document demonstration; 20 constraints and persona review; 15 prompt critique
11:15–11:30	Tea break	15 minutes
11:30–13:00	3 · Engineering methodology and Excel	15 primer; 25 guided SPC; 20 guided FMEA; 20 guided DMAIC; 10 shared review
13:00–14:00	Lunch	60 minutes
14:00–15:15	4 · Copilot and reasoning agents	15 concepts; 25 FMEA/report assistant demo; 20 research/analysis comparison; 15 questions and buffer
15:15–15:30	Tea break	15 minutes
15:30–16:15	5 · SharePoint grounding	10 grounding; 20 retrieval/citation demo; 15 permissions and discussion
16:15–16:45	6 · AI security and toolkit	20 seven-step review; 10 personal prompt toolkit
16:45–17:00	Closing and Q&A	10 recap; 5 next action

Only three short guided exercises

Use the separate Starter.xlsx in each lab. Each workbook contains one small source table and one prompt. Open Worked-Reference.xlsx only after the learner has checked the Copilot draft.

1 · Copilot ecosystem

Show the same engineering question in Chat and Excel. Identify which evidence is available in each surface. Complete the access check once; use a paired screen or the supplied worked reference when access cannot be resolved within ten minutes.

2 · Prompting for engineering documents

Teach RISE: Role, Input, Steps, Expectations; E-MAP: Engineering context, Metrics, Assumptions, Parameters. Demonstrate converting one case note into a short specification and report. Compare a vague prompt with an evidence-constrained prompt. Demonstrate reliability, operations and safety perspectives, one challenge per role, and a moderator summary preserving disagreement. Participants critique the displayed prompt; no separate document pack is required.

4 · Copilot and reasoning agents

Trainer demonstrates a prepared FMEA/report assistant: bounded instructions, rating-scale knowledge, one test and one correction. Compare a cited research response with an analysis response using the case evidence. Discuss the minutes-to-actions-to-report pattern. No learner creates, deploys or shares an agent. If a tenant feature is unavailable, walk through the prepared instructions and worked example; identify this as a demonstration fallback.

5 · SharePoint grounding

Trainer shows a prepared approved library, a question, retrieved passage and cited answer. Open the citation and check revision, scope and permissions. Discuss how a grounded answer could support an engineering report. No learner creates a site, library or agent. If live access is unavailable, use a labelled document-retrieval walkthrough rather than claiming a live agent run.

6 · AI security and toolkit

Apply seven checks to one displayed workbook claim: classification; governance/access; secure prompting; output verification; human sign-off; traceability/audit; awareness and incident reporting. Treat instructions embedded in retrieved documents as data, not authority. Learners retain one reusable prompt and one review rule; no additional lab submission.

Module 3 · SPC | 25 minutes

5 min · Read the prepared analysis

Open Lab-01-SPC/Starter.xlsx. Review the 12 rows in Pump_Readings: eight Baseline and four Monitoring. Select the named table PumpReadings before opening Copilot.

12 min · Prompt and inspect

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

8 min · Verify and retain

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

Core prompt · Plot and interpret

Use only PumpReadings. Calculate the baseline mean, sample standard deviation, UCL and LCL; flag the four monitoring rows; and create a line chart. Verify T10 against the LCL and state that a signal does not prove cause.

Module 3 · FMEA | 20 minutes

4 min · Read the prepared FMEA

Open Lab-02-FMEA/Starter.xlsx. Review the four supplied failure modes and ratings in FMEA_Input. Use Copilot to create the formula-based output; do not invent or rescore rows.

10 min · Prompt and check priorities

Paste the one prompt from Copilot_Prompt. Ask Copilot to calculate RPN, rank the four rows and flag Severity 9 or higher. Independently verify FM01 and FM04.

6 min · Review one row

Write one evidence gap for FM01 and one action proposal. Keep residual ratings unassessed. The multi-persona debate is a trainer demonstration in Module 2, not an extra participant task.

Core prompt · Calculate and prioritise

Use only FMEAInput. Preserve the four supplied rows and ratings; calculate RPN, rank the rows and flag Severity 9 or higher. Verify FM01 = 140 and FM04 Severity = 9.

Module 3 · DMAIC | 20 minutes

4 min · Read the prepared rates

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

10 min · Prompt a short summary

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

6 min · State a decision boundary

Save three cells: measured change, limitation and next evidence. The absolute change is 5.95238 percentage points. Two concurrent changes and a short pilot do not prove a cause or sustained improvement.

Core prompt · Define and measure

Open Lab-03-DMAIC/Starter.xlsx. Review the six weekly records in Weekly_Data. Use Copilot to calculate phase rates and draft the five-row DMAIC summary.

Outcome coverage

LO1 methodology: primer and guided Excel review. LO2 documentation: trainer examples and Excel outputs; no independent Word/PPT pack. LO3 prompting: framework, critique and three selected prompts. LO4–LO6 agents, reasoning and SharePoint: demonstration-level coverage only. LO7 security: review of the displayed case. Independent agent deployment is not a completion claim for this shortened delivery.

Presentation teaching sequence — v3.1

Module 1 introduces AI, Copilot and the pump case before workbook names. Module 2 moves from a vague prompt to RISE, E-MAP, constraints and a worked persona challenge. Module 3 explains each method before its guided exercise. Modules 4–5 explain agents and grounding before demonstrations. Module 6 applies security checks and closes. The original clock times and 65-minute exercise total are unchanged.

Using the expanded 206-slide deck

Slides 1–79 are the one-day teaching route. Slide 80 introduces optional reference material. Slides 81–200 contain 30 worked examples; they are not additional scheduled labs. Keep the original six-module timetable and the 25/20/20-minute guided Excel exercises.

Reference index: SPC slides 87-118; FMEA 119-142; DMAIC 143-166; Prompting 167-186; Agents 187-194; Grounding 195-202; Security 203-206.

Each reference example follows data, prompt, illustrative answer and verification. Supporting CSVs, prompts and Worked-Examples.pdf are in Reference-Examples. The trainer demonstrations use the supplied Trainer-Demonstrations pack.

LO2 practical workflow: create the engineering output with Copilot 365

Learners repeat the same five actions in all three exercises: open the small Starter workbook; select the named table; open Copilot in Excel; paste the one supplied prompt; inspect and independently verify one result. The Worked-Reference workbook is opened only after the attempt.

Lab 1 - Use Copilot in Excel to create an SPC review (25 minutes)

Source: Pump_Readings / PumpReadings.

Copilot prompt: Act as an engineering data assistant. Use only the named Excel table PumpReadings. Use the eight Baseline rows to calculate the mean, sample standard deviation, UCL = mean + 3 x standard deviation, and LCL = mean - 3 x standard deviation. Flag each Monitoring row as Below LCL, Above UCL, or Within limits. Create a line chart of all 12 readings with the centre line, UCL and LCL. Return a three-sentence interpretation. Do not claim a root cause or that the pump is safe. Show the formulas used.

Independent check: Verify T10 = 59.4 L/h against the calculated LCL. A point below the LCL is a signal to investigate, not a diagnosis.

Lab 2 - Use Copilot in Excel to create an FMEA review (20 minutes)

Source: FMEA_Input / FMEAInput.

Copilot prompt: Act as an FMEA facilitator. Use only the named Excel table FMEAInput and preserve every row and supplied rating. Create an FMEA output table, calculate RPN = Severity x Occurrence x Detection, rank the four rows from highest to lowest RPN, and flag any row with Severity 9 or higher. For each row, propose one evidence request and one practical follow-up action. Do not invent incidents, change ratings, or approve the design. Show the RPN formula.

Independent check: Verify FM01: 7 x 5 x 4 = 140. Also verify that FM04 is flagged because Severity = 9 even though its RPN is lower.

Lab 3 - Use Copilot in Excel to create a DMAIC summary (20 minutes)

Source: Weekly_Data / WeeklyData.

Copilot prompt: Act as a Six Sigma project assistant. Use only the named Excel table WeeklyData. Calculate total opportunities, total deviations, and deviation rate for Baseline and Pilot using deviations divided by opportunities. Create a five-row DMAIC summary: Define, Measure, Analyse, Improve, Control. Clearly separate observed facts from hypotheses. State that the short pilot does not prove the cause of the change. Show the formulas and return one next evidence check.

Independent check: Verify Baseline = 36 / 400 = 9.0% and Pilot = 7 / 200 = 3.5%. The lower pilot rate is an observation, not causal proof.

Microsoft 365 Copilot links

Microsoft 365 Copilot: https://m365.cloud.microsoft/chat

Get started with Copilot in Excel: https://support.microsoft.com/en-us/office/get-started-with-copilot-in-excel

How Copilot Chat works in Microsoft 365 apps: https://support.microsoft.com/en-us/microsoft-365-copilot/how-copilot-chat-works-in-microsoft-365-apps